

CHAPTER 6: SECURITY POLICIES AND GOVERNANCE

Learning Outcome Focus: LO1, LO4 - Explain security concepts and the impact of cybersecurity on civil and military institutions, privacy, business and government applications.

6.1 Security Policies and Standards

Security policies are the foundation of an organization's cybersecurity program. They are high-level documents that outline the organization's security goals, objectives, and requirements. Security policies are typically developed by senior management and are designed to be technology-neutral.

- ✓ • **Components of a Security Policy:** A typical security policy will include the following components:
 - **Purpose:** A statement of the policy's purpose and scope.
 - **Roles and Responsibilities:** A description of the roles and responsibilities of individuals and groups within the organization.
 - **Policy Statements:** A set of high-level statements that outline the organization's security requirements.
 - **Compliance:** A description of how compliance with the policy will be monitored and enforced.
- ✓ • **Security Standards:** Security standards are more detailed documents that provide specific guidance on how to implement the security policies. They are often based on industry best practices and may be technology-specific.

6.2 Information Security Management Systems (ISMS)

An **Information Security Management System (ISMS)** is a systematic approach to managing an organization's information security. An ISMS is based on a cycle of planning, implementing, reviewing, and improving the organization's security controls.

- **ISO/IEC 27001 and 27002:** These are international standards that provide a framework for establishing, implementing, maintaining, and continually improving an ISMS.
- **NIST Cybersecurity Framework:** This is a voluntary framework that provides a set of guidelines for managing cybersecurity risk. It is widely used by organizations in the United States and around the world.
- **CIS Critical Security Controls:** This is a prioritized set of actions that organizations can take to protect themselves against the most common cyber attacks.

ISO - International organization for Standardization
IEC - International Electrotechnical Commission

6.3 Risk Management

Risk management is the process of identifying, assessing, and mitigating risks to an organization's assets. It is a critical component of any cybersecurity program.

- **Risk Identification:** The process of identifying potential risks to the organization's assets.
- **Risk Assessment:** The process of assessing the likelihood and impact of each identified risk.
- **Risk Mitigation:** The process of implementing controls to reduce the likelihood or impact of each identified risk.

6.4 Incident Response Planning

Incident response planning is the process of developing a plan for responding to security incidents. An incident response plan should be designed to minimize the impact of a security incident and to ensure that the organization can recover as quickly as possible.

- **Incident Response Team:** An incident response plan should identify the members of the incident response team and their roles and responsibilities.
- **Incident Response Procedures:** An incident response plan should include a set of procedures for responding to different types of security incidents.

6.5 Disaster Recovery and Business Continuity

Disaster recovery is the process of recovering from a major disruption to the organization's IT systems. **Business continuity** is the process of ensuring that the organization can continue to operate in the event of a major disruption.

- **Recovery Time Objective (RTO):** The maximum amount of time that a system can be down after a disaster.
- **Recovery Point Objective (RPO):** The maximum amount of data that can be lost after a disaster.

6.6 Chapter Summary and Review Questions

This chapter has provided an overview of security policies and governance. We have discussed the importance of security policies and standards, the different types of information security management systems, the process of risk management, and the importance of incident response planning and disaster recovery. A thorough understanding of these concepts is essential for developing effective cybersecurity strategies and for protecting against the ever-present threat of cyber-attacks.

Review Questions:

- 1 What is the purpose of a security policy?
- 2 What is an Information Security Management System (ISMS)?
- 3 What are the three main steps in the risk management process?
- 4 What is the purpose of an incident response plan?
- 5 What is the difference between disaster recovery and business continuity?

References:

- [1] International Organization for Standardization. (2013). *ISO/IEC 27001:2013: Information technology — Security techniques — Information security management systems — Requirements*. <https://www.iso.org/standard/54534.html>
- [2] National Institute of Standards and Technology. (2018). *Framework for Improving Critical Infrastructure Cybersecurity, Version 1.1*. <https://nvlpubs.nist.gov/nistpubs/CSWP/NIST.CSWP.04162018.pdf>

CHAPTER 7: NETWORK AND SYSTEM SECURITY

Learning Outcome Focus: LO3 - Apply the techniques for identifying, detecting, and defending against cybersecurity threats, attacks and protecting information assets.

7.1 Network Security Fundamentals

Network security is the practice of preventing and protecting against unauthorized intrusion into corporate networks. It involves the use of hardware and software to protect the usability and integrity of your network and data. Effective network security manages access to the network. It targets a variety of threats and stops them from entering or spreading on your network.

- **Network Segmentation:** This is the practice of dividing a computer network into smaller, isolated subnetworks. This can help to contain the spread of malware and to limit the impact of a security breach.
- **Firewalls:** A firewall is a network security device that monitors incoming and outgoing network traffic and decides whether to allow or block specific traffic based on a defined set of security rules.
- **Network Access Control (NAC):** NAC is a security solution that enforces policy on devices that access networks to increase network visibility and reduce risk.

7.2 Intrusion Detection and Prevention

Intrusion Detection Systems (IDS) and **Intrusion Prevention Systems (IPS)** are network security technologies that are used to detect and prevent malicious activity.

- **IDS:** An IDS is a passive system that monitors network traffic and alerts administrators when it detects suspicious activity.
- **IPS:** An IPS is an active system that is placed in-line with network traffic and can block malicious traffic in real-time.
- **Network-Based vs. Host-Based:** An IDS/IPS can be either network-based or host-based. A network-based IDS/IPS monitors traffic on the entire network, while a host-based IDS/IPS monitors activity on a single host.
- **Signature-Based vs. Anomaly-Based:** An IDS/IPS can use either signature-based or anomaly-based detection. Signature-based detection looks for known patterns of malicious activity, while anomaly-based detection looks for deviations from normal behavior.

7.3 Operating System Security

Operating system (OS) security is the process of ensuring the integrity, confidentiality, and availability of an operating system. OS security refers to specified steps or measures used to protect the OS from threats, viruses, worms, malware, or remote hacker intrusions.

- **OS Hardening:** This is the process of configuring an OS to be more secure. This can involve disabling unnecessary services, changing default passwords, and applying security patches.
- **Patch Management:** This is the process of applying updates to software to fix security vulnerabilities.

7.4 Endpoint Protection

Endpoint protection is the practice of securing endpoints or entry points of end-user devices such as desktops, laptops, and mobile devices from being exploited by malicious actors and campaigns.

- **Antivirus and Anti-Malware:** This is software that is designed to detect and remove malware.
- **Endpoint Detection and Response (EDR):** EDR is a security solution that continuously monitors and responds to advanced threats.

7.5 Secure Communications

Secure communications is the practice of encrypting data in transit to protect it from eavesdropping and tampering.

- **Virtual Private Networks (VPNs):** A VPN is a technology that creates a secure, encrypted connection over a less secure network, such as the internet.
- **Secure Shell (SSH):** SSH is a cryptographic network protocol for operating network services securely over an unsecured network.

7.6 Chapter Summary and Review Questions

This chapter has provided an overview of network and system security. We have discussed the fundamentals of network security, the different types of intrusion detection and prevention systems, the importance of operating system security and endpoint protection, and the various technologies that can be used to secure communications. A thorough understanding of these concepts is essential for developing effective cybersecurity strategies and for protecting against the ever-present threat of cyber-attacks.

Review Questions:

- 1 What is the purpose of network segmentation?
- 2 What is the difference between an IDS and an IPS?
- 3 What is OS hardening?
- 4 What is the purpose of an EDR solution?
- 5 What is a VPN and how does it work?

References:

- [1] Tanenbaum, A. S., & Wetherall, D. J. (2011). *Computer Networks*. Pearson.
- [2] Stallings, W. (2017). *Cryptography and Network Security: Principles and Practice*. Pearson.

CHAPTER 8: SOFTWARE SECURITY AND VULNERABILITIES

Learning Outcome Focus: LO3, LO5 - Apply the techniques for identifying, detecting, and defending against cybersecurity threats, and identify the methods and motives of cybersecurity incident perpetrators, and the countermeasures employed by organisations and agencies to prevent and detect those incidences and software application vulnerabilities.

8.1 Software Development Security

Software development security is the process of integrating security into the software development lifecycle (SDLC). The goal of software development security is to build secure software from the ground up, rather than trying to add security on as an afterthought.

- **Secure Software Development Lifecycle (SDLC):** A secure SDLC is a software development process that includes security-related activities at each stage of the development process. This can include threat modeling, secure coding, and security testing.
- **Threat Modeling:** This is the process of identifying potential threats to a software application and designing countermeasures to mitigate those threats.
- **Secure Coding:** This is the practice of writing code that is resistant to attack. This can include using secure coding libraries, following secure coding best practices, and performing code reviews.

8.2 Common Software Vulnerabilities

There are a variety of common software vulnerabilities that can be exploited by attackers. The **OWASP Top 10** is a list of the most critical web application security risks.

- **Injection:** This is a type of vulnerability that allows an attacker to inject malicious code into a software application.
- **Broken Authentication:** This is a type of vulnerability that allows an attacker to bypass a software application's authentication mechanisms.
- **Sensitive Data Exposure:** This is a type of vulnerability that allows an attacker to access sensitive data, such as passwords and credit card numbers.
- **XML External Entities (XXE):** This is a type of vulnerability that allows an attacker to interfere with an application's processing of XML data.
- **Broken Access Control:** This is a type of vulnerability that allows an attacker to access resources that they are not authorized to access.

8.3 Vulnerability Management

Vulnerability management is the process of identifying, assessing, and mitigating vulnerabilities in software applications and systems.

- **Vulnerability Scanning:** This is the process of using automated tools to scan for vulnerabilities.
- **Vulnerability Assessment:** This is the process of assessing the severity of each identified vulnerability.
- **Vulnerability Mitigation:** This is the process of implementing controls to reduce the risk of each identified vulnerability.

8.4 Application Security Testing

Application security testing is the process of testing a software application for security vulnerabilities.

- **Static Application Security Testing (SAST):** This is a type of testing that analyzes the source code of a software application for security vulnerabilities.
- **Dynamic Application Security Testing (DAST):** This is a type of testing that analyzes a running software application for security vulnerabilities.
- **Interactive Application Security Testing (IAST):** This is a type of testing that combines the features of SAST and DAST.

8.5 Chapter Summary and Review Questions

This chapter has provided an overview of software security and vulnerabilities. We have discussed the importance of software development security, the different types of common software vulnerabilities, the process of vulnerability management, and the various types of application security testing. A thorough understanding of these concepts is essential for developing effective cybersecurity strategies and for protecting against the ever-present threat of cyber-attacks.

Review Questions:

- 1 What is the purpose of a secure SDLC?
- 2 What is threat modeling?
- 3 What are the three main types of application security testing?
- 4 What is the OWASP Top 10?
- 5 What is the difference between vulnerability scanning and vulnerability assessment?

References:

- [1] OWASP Foundation. *OWASP Top Ten*. <https://owasp.org/www-project-top-ten/>
- [2] McGraw, G. (2006). *Software Security: Building Security In*. Addison-Wesley Professional.

CHAPTER 9: INCIDENT RESPONSE AND MANAGEMENT

Learning Outcome Focus: LO3, LO5 - Apply the techniques for identifying, detecting, and defending against cybersecurity threats, and identify the methods and motives of cybersecurity incident perpetrators, and the countermeasures employed by organisations and agencies to prevent and detect those incidences.

9.1 Incident Response Framework

Incident response is the process of responding to and managing a security incident. The goal of incident response is to minimize the impact of a security incident and to restore normal operations as quickly as possible.

- **Incident Response Lifecycle:** The incident response lifecycle is a four-phase process that is used to manage security incidents:
 - **Preparation:** This phase involves preparing the organization to respond to security incidents. This can include developing an incident response plan, training the incident response team, and acquiring the necessary tools and resources.
 - **Detection and Analysis:** This phase involves detecting and analyzing security incidents. This can include monitoring security logs, using intrusion detection systems, and analyzing suspicious activity.
 - **Containment, Eradication, and Recovery:** This phase involves containing the security incident, eradicating the cause of the incident, and recovering from the incident.
 - **Post-Incident Activity:** This phase involves reviewing the incident and identifying lessons learned. This can help to improve the organization's incident response capabilities.

9.2 Incident Detection and Analysis

Incident detection is the process of identifying that a security incident has occurred. **Incident analysis** is the process of determining the scope and impact of a security incident.

- **Security Monitoring:** This is the process of monitoring security logs and other data sources for signs of malicious activity.
- **Log Analysis:** This is the process of analyzing security logs to identify suspicious activity.

- **Indicators of Compromise (IoCs):** These are pieces of forensic data that can be used to identify a security incident.

9.3 Incident Containment and Eradication

Incident containment is the process of preventing a security incident from spreading. **Incident eradication** is the process of removing the cause of a security incident.

- **Short-Term Containment:** This involves taking immediate steps to contain the security incident, such as disconnecting a compromised system from the network.
- **Long-Term Containment:** This involves taking more permanent steps to contain the security incident, such as rebuilding a compromised system.

9.4 Incident Recovery and Lessons Learned

Incident recovery is the process of restoring normal operations after a security incident. **Lessons learned** is the process of reviewing a security incident and identifying ways to improve the organization's incident response capabilities.

- **System Recovery:** This involves restoring systems and data from backups.
- **Post-Incident Review:** This is a meeting that is held after a security incident to review the incident and identify lessons learned.

9.5 Chapter Summary and Review Questions

This chapter has provided an overview of incident response and management. We have discussed the incident response lifecycle, the different phases of incident response, and the importance of incident detection, analysis, containment, eradication, and recovery. A thorough understanding of these concepts is essential for developing effective cybersecurity strategies and for protecting against the ever-present threat of cyber attacks.

Review Questions:

- 1 What are the four phases of the incident response lifecycle?
- 2 What is the purpose of incident detection and analysis?
- 3 What is the difference between short-term and long-term containment?
- 4 What is the purpose of a post-incident review?
- 5 What are some of the key challenges in incident response?

References:

- [1].National Institute of Standards and Technology (NIST). (2012). *Computer Security Incident Handling Guide*. <https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-61r2.pdf>
- [2] Cichonski, P., Millar, T., Grance, T., & Scarfone, K. (2012). *Guide to Cyber Threat Information Sharing*. National Institute of Standards and Technology. <https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-150.pdf>

CHAPTER 10: THREAT INTELLIGENCE AND HUNTING

Learning Outcome Focus: LO2, LO5 - Describe common cyber-attacks and threats, and identify the methods and motives of cybersecurity incident perpetrators, and the countermeasures employed by organisations and agencies to prevent and detect those incidences.

10.1 Threat Intelligence Fundamentals

Threat intelligence is evidence-based knowledge, including context, mechanisms, indicators, implications, and actionable advice, about an existing or emerging menace or hazard to assets that can be used to inform decisions regarding the subject's response to that menace or hazard. The goal of threat intelligence is to provide organizations with the information they need to make better security decisions.

- **Threat Intelligence Lifecycle:** The threat intelligence lifecycle is a six-phase process that is used to produce threat intelligence:
 - **Planning and Direction:** This phase involves defining the goals and objectives of the threat intelligence program.
 - **Collection:** This phase involves collecting raw data from a variety of sources.
 - **Processing:** This phase involves converting the raw data into a usable format.
 - **Analysis:** This phase involves analyzing the processed data to identify threats and patterns.
 - **Dissemination:** This phase involves disseminating the threat intelligence to stakeholders.
 - **Feedback:** This phase involves getting feedback from stakeholders to improve the threat intelligence program.

10.2 Threat Actor Profiling

Threat actor profiling is the process of identifying and analyzing the characteristics of threat actors. This can include their motivations, capabilities, and tactics, techniques, and procedures (TTPs).

- **Cybercriminals:** These are individuals or groups who are motivated by financial gain.
- **Nation-States:** These are government-sponsored groups who are motivated by political or economic interests.
- **Hacktivists:** These are individuals or groups who are motivated by a political or social cause.

10.3 Indicators of Compromise (IoCs)

Indicators of Compromise (IoCs) are pieces of forensic data that can be used to identify a security incident. IoCs can include things like IP addresses, domain names, file hashes, and registry keys.

10.4 Threat Hunting

Threat hunting is the proactive process of searching for cyber threats that are lurking undetected in a network. Threat hunting is a more proactive approach to security than traditional security monitoring, which is focused on detecting known threats.

- **Hypothesis-Driven Hunting:** This is a type of threat hunting where the hunter starts with a hypothesis about a potential threat and then searches for evidence to support or refute that hypothesis.
- **Data-Driven Hunting:** This is a type of threat hunting where the hunter starts with a large dataset and then uses data analysis techniques to identify suspicious activity.

10.5 Chapter Summary and Review Questions

This chapter has provided an overview of threat intelligence and hunting. We have discussed the threat intelligence lifecycle, the different types of threat actors, the importance of IoCs, and the different approaches to threat hunting. A thorough understanding of these concepts is essential for developing effective cybersecurity strategies and for protecting against the ever-present threat of cyber-attacks.

Review Questions:

- 6 What are the six phases of the threat intelligence lifecycle?
- 7 What are the three main types of threat actors?
- 8 What is an Indicator of Compromise (IoC)?

- 9 What is the difference between hypothesis-driven and data-driven *threat hunting*?
- 10 What are some of the key challenges in threat intelligence?

References:

- [1] Shackleford, D. (2015). *Threat Intelligence: The Practitioner's Guide to Threat Intelligence*. SANS Institute.
- [2] The MITRE Corporation. (2021). *ATT&CK*. <https://attack.mitre.org/>

CHAPTER 11: CYBERSECURITY STRATEGY AND POLICY

Learning Outcome Focus: LO4, LO6 - Explain the impact of cybersecurity on civil and military institutions, privacy, business and government applications, and evaluate cybersecurity and national security strategies to the typologies of cyber-attacks that require policy tools and domestic response.

11.1 Cybersecurity Strategy Development

A **cybersecurity strategy** is a high-level plan that outlines how an organization will protect itself from cyber threats. A cybersecurity strategy should be aligned with the organization's overall business goals and objectives.

- **Strategic Planning Process:** The strategic planning process for cybersecurity should include the following steps:
 - **Identify Assets:** The first step is to identify the organization's critical assets, such as data, systems, and intellectual property.
 - **Assess Risks:** The next step is to assess the risks to the organization's assets. This should include identifying potential threats and vulnerabilities.
 - **Develop Controls:** The next step is to develop controls to mitigate the identified risks.
 - **Implement Controls:** The next step is to implement the developed controls.
 - **Monitor and Review:** The final step is to monitor and review the effectiveness of the controls.

11.2 National Cybersecurity Strategy

A **national cybersecurity strategy** is a plan that outlines how a country will protect itself from cyber threats. A national cybersecurity strategy should be developed in consultation with a variety of stakeholders, including government, industry, and academia.

- **Critical Infrastructure Protection:** A national cybersecurity strategy should include a plan for protecting the country's critical infrastructure, such as power grids, financial systems, and transportation networks.
- **Public-Private Partnerships:** A national cybersecurity strategy should promote public-private partnerships to share information and resources.
- **International Cooperation:** A national cybersecurity strategy should promote international cooperation to address the global nature of cyber threats.

11.3 Cybersecurity and National Security

Cybersecurity is a critical component of national security. Cyber attacks can be used to disrupt critical infrastructure, steal sensitive information, and influence public opinion.

- **Cyber Warfare:** Cyber warfare is the use of cyber attacks as part of a military campaign.
- **Espionage:** Cyber espionage is the use of cyber attacks to steal sensitive government or corporate secrets.
- **Critical Infrastructure Resilience:** This is the ability of critical infrastructure to withstand and recover from a cyber attack.

11.4 Cybersecurity in Different Sectors

Cybersecurity is important in all sectors of the economy. However, some sectors are more critical than others.

- **Financial Sector:** The financial sector is a prime target for cyber attacks due to the large amount of money that is at stake.
- **Healthcare Sector:** The healthcare sector is a prime target for cyber attacks due to the sensitive nature of the data that it holds.
- **Energy Sector:** The energy sector is a prime target for cyber attacks due to the critical role that it plays in the economy.

11.5 Chapter Summary and Review Questions

This chapter has provided an overview of cybersecurity strategy and policy. We have discussed the importance of developing a cybersecurity strategy, the key components of a national cybersecurity strategy, the relationship between cybersecurity and national security, and the importance of cybersecurity in different sectors of the economy. A thorough understanding of

these concepts is essential for developing effective cybersecurity strategies and for protecting against the ever-present threat of cyber-attacks.

Review Questions:

- 1 What are the five steps in the strategic planning process for cybersecurity?
- 2 What are the three key components of a national cybersecurity strategy?
- 3 What is the difference between cyber warfare and cyber espionage?
- 4 Why is the financial sector a prime target for cyber-attacks?
- 5 What are some of the key challenges in developing a national cybersecurity strategy?

References:

- [1] The White House. (2023). *National Cybersecurity Strategy*. <https://www.whitehouse.gov/wp-content/uploads/2023/03/National-Cybersecurity-Strategy-2023.pdf>
- [2] Cybersecurity and Infrastructure Security Agency (CISA). (2021). *Critical Infrastructure Sectors*. <https://www.cisa.gov/critical-infrastructure-sectors>

CHAPTER 12: CYBERSECURITY ETHICS AND PROFESSIONAL RESPONSIBILITY

Learning Outcome Focus: LO6 - State the ethical obligations of security professionals, evaluate cybersecurity and national security strategies to the typologies of cyber-attacks that require policy tools and domestic response.

12.1 Cybersecurity Ethics Framework

Cybersecurity ethics is a branch of ethics that focuses on the moral and ethical issues that arise in the field of cybersecurity. It is a critical component of any cybersecurity program, as it provides a framework for making decisions that are not only technically sound but also morally and ethically responsible.

- **Ethical Principles in Cybersecurity:** There are a number of ethical principles that are relevant to cybersecurity, including:
 - **Beneficence:** The principle of doing good and avoiding harm.

- **Non-maleficence:** The principle of not causing harm.
- **Justice:** The principle of fairness and equity.
- **Respect for persons:** The principle of respecting the autonomy and dignity of individuals.
- **Professional Codes of Ethics:** There are a number of professional codes of ethics that provide guidance for cybersecurity professionals, including:
 - **(ISC)² Code of Ethics:** This code of ethics is for certified information systems security professionals.
 - **ACM Code of Ethics:** This code of ethics is for computing professionals.
 - **ISSA Code of Ethics:** This code of ethics is for information systems security association members.

12.2 Ethical Obligations of Security Professionals

Security professionals have a number of ethical obligations, including:

- **Protecting society and the common good:** Security professionals have a responsibility to protect society and the common good from the harms of cyber attacks.
- **Acting with honesty and integrity:** Security professionals have a responsibility to act with honesty and integrity in all of their professional dealings.
- **Providing competent and diligent service:** Security professionals have a responsibility to provide competent and diligent service to their employers and clients.
- **Advancing the profession:** Security professionals have a responsibility to advance the profession of cybersecurity by sharing their knowledge and expertise.

12.3 Ethical Challenges in Cybersecurity

There are a number of ethical challenges that arise in the field of cybersecurity, including:

- **Surveillance and privacy:** The use of surveillance technologies can raise privacy concerns.
- **Ethical hacking and penetration testing:** The use of ethical hacking and penetration testing can raise ethical concerns if it is not done in a responsible manner.
- **Responsible disclosure:** The process of disclosing security vulnerabilities can raise ethical concerns if it is not done in a way that protects the public.

12.4 Privacy and Data Protection

Privacy is the right of individuals to control their personal information. **Data protection** is the process of protecting personal information from unauthorized access, use, or disclosure.

- **Privacy Principles:** There are a number of privacy principles that are relevant to cybersecurity, including:
 - **Collection limitation:** The principle that personal information should only be collected for a legitimate purpose.
 - **Data quality:** The principle that personal information should be accurate, complete, and up-to-date.
 - **Purpose specification:** The principle that the purpose for which personal information is collected should be specified at the time of collection.
 - **Use limitation:** The principle that personal information should only be used for the purpose for which it was collected.

12.5 Chapter Summary and Review Questions

This chapter has provided an overview of cybersecurity ethics and professional responsibility. We have discussed the importance of a cybersecurity ethics framework, the ethical obligations of security professionals, the ethical challenges that arise in the field of cybersecurity, and the importance of privacy and data protection. A thorough understanding of these concepts is essential for developing effective cybersecurity strategies and for protecting against the ever-present threat of cyber attacks.

Review Questions:

- 1 What are the four main ethical principles in cybersecurity?
- 2 What are the four main ethical obligations of security professionals?
- 3 What are some of the ethical challenges that arise in the field of cybersecurity?
- 4 What are the four main privacy principles?
- 5 What is the difference between privacy and data protection?

References:

- [1] (ISC)². *ISC2 Code of Ethics*. <https://www.isc2.org/ethics>
- [2] Association for Computing Machinery (ACM). *ACM Code of Ethics and Professional Conduct*. <https://www.acm.org/code-of-ethics>